EconLab Classroom

A Modular Toolkit for Teaching Economics Through Experiments

Matteo Ploner

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Abstract

This paper presents *EconLab Classroom*, a modular toolkit for teaching economics through interactive classroom experiments. The project translates economic environments into browser-based oTree applications that can be used during lectures, tutorials, and laboratory sessions. Its main objective is to help students connect formal models with observed behavior by first participating in decision tasks and then analyzing the data produced by the class. The paper describes the pedagogical rationale of the project, the organization of the modules, the technical infrastructure supporting classroom use, and a detailed implementation of the Beauty Contest game. The discussion highlights how real-time data collection, anonymized repeated participation, instructor-facing reports, and AI-mediated feedback tools can turn ordinary teaching sessions into empirical exercises in which theoretical predictions, behavioral heterogeneity, and institutional design are examined together.

Introduction

Economic experiments have a long history in teaching. One useful point of departure is Vernon Smith's encounter with Chamberlin's classroom market experiment at Harvard. In Bergstrom's account, Smith recalled: "I did observe and participate in Chamberlin's little 'experiment'." (Bergstrom, 2003). Economic principles are generally taught with reliance on verbal explanations, mathematical models, and static diagrams. This warrants a high level of precision and rigor, but it can be difficult for students to grasp the intuition behind the concepts and to connect them to real-world phenomena. This point is central in Vernon Smith's account of experimental economics: economic training has often emphasized theory more than observation, while experimentation changes the discipline by making economic propositions

empirically demonstrable and replicable. In this view, experiments do not replace theory; rather, they provide a dynamic and interactive way of learning economics by allowing students to experience economic principles in action and to confront theoretical claims with observed behavior (Smith, 1989).

Classroom experiments make this connection pedagogically productive. Students formulate expectations, act under explicit rules and incentives, and then interpret a dataset that includes their own decisions. This sequence is central to Bergstrom and Miller's account of classroom experiments: the experiment comes before the explanation, so theory becomes a tool for interpreting outcomes that students have just generated rather than material to be memorized in isolation (Bergstrom and Miller, 1997). The argument is especially relevant for students accustomed to interactive and digital environments, for whom active participation can make abstract economic concepts more concrete and easier to discuss (Picault, 2019).

Traditionally, economic experiments have been conducted in physical classrooms using physical materials (e.g., cards, coins, tokens) or computer software (Holt, 2007). However, the advent of online platforms and tools has made it possible to conduct economic experiments in virtual classrooms, allowing for greater flexibility and accessibility. The use of online platforms also allows data to be collected in real time and used for analysis and discussion in class.

The contribution of this paper is threefold. First, it presents *EconLab Classroom* as a modular, open, and adaptable suite of oTree applications for teaching economics through classroom experiments. Second, it documents the pedagogical logic and technical infrastructure that make these applications usable in ordinary teaching settings, including browser-based participation, anonymized longitudinal identifiers, real-time data collection, instructor-facing reports, and optional AI-mediated feedback for selected apps. Third, it illustrates the approach through a detailed Beauty Contest application, showing how a canonical strategic interaction can be translated into a classroom activity that links theory, implementation, observed behavior, and post-experimental discussion.

Description

EconLab Classroom is a collection of interactive economic experiments designed mainly for university teaching and implemented using the oTree framework (Chen et al., 2016). Its main purpose is to strengthen the connection between formal theory and observed behavior by moving part of the learning process from exposition to participation.

The applications are intended to be used during lectures, tutorials, and laboratory sessions. Each app implements a specific economic environment and produces data that can be discussed immediately after the activity. This makes the classroom a small-scale empirical setting: students first encounter incentives as decision makers and then return to the theoretical framework as analysts of the resulting data.

Overview of the EconLab Classroom Apps

The apps are organized around three broad teaching objectives. First, they make abstract concepts operational by asking students to act within the formal environment described by the theory. Second, they highlight behavioral heterogeneity and systematic deviations from benchmark assumptions, such as risk neutrality or purely self-interested preferences. Third, they generate data that can be used to discuss incentives, efficiency, distributional outcomes, and the role of institutions.

As illustrated in Table 1, the structure of the project covers three main areas in which classroom experiments are especially useful: individual preferences and traits, strategic interaction, and market or auction institutions. The modules can be used independently, but they can also be combined into longer sequences so that students encounter related ideas across different institutional settings.

Table 1: Overview of EconLab Classroom apps.

Area	Example modules	Main concepts	Classroom output
Preferences and traits	MPL, CRT, confidence tasks, heuristics tasks	Risk preferences, cognitive reflection, beliefs, confidence, behavioral biases	Individual-level measures, distributions of choices, comparisons across tasks
Strategic interaction	Beauty Contest, Prisoner's Dilemma, public good games, dictator games	Best responses, equilibrium, cooperation, social preferences, iterated reasoning	Group outcomes, payoff patterns, distributions of strategies and beliefs, debriefing prompts
Markets and auctions	Continuous double auctions, sealed-bid mechanisms, BDM procedures	Price formation, gains from trade, allocative efficiency, mechanism design	Transaction prices, allocations, surplus measures, efficiency indicators

Preferences & Traits

This group of applications is designed to elicit individual characteristics and behavioral parameters. These modules cover domains such as risk preferences, time preferences, social preferences, beliefs, and confidence. Their pedagogical role is to make heterogeneity visible. Rather than treating preferences as abstract primitives only, students observe how individual differences can be measured and how these differences affect economic choices.

These activities are particularly useful when introducing utility, expected utility, intertemporal choice, altruism, inequality aversion, and belief formation. They also create a natural bridge between theoretical models and empirical measurement, since the data collected in class can be summarized, compared, and interpreted with the tools introduced in the course.

Strategic Interaction

The second group of applications focuses on strategic environments. These include canonical games such as the Prisoner's Dilemma and the Beauty Contest, as well as competition games and other normal-form or dynamic interactions. The purpose is to let students experience strategic interdependence directly: each participant's outcome depends not only on their own action, but also on the actions, expectations, and reasoning of others.

These modules support the teaching of equilibrium concepts, best responses, iterated reasoning, coordination, cooperation, and learning. They are also useful for discussing why observed play may differ from benchmark predictions and how repeated interaction, feedback, and institutional details shape behavior.

Markets and Auctions

The third group of applications implements market and auction institutions. Examples include continuous double auctions, sealed-bid auctions, and mechanisms such as BDM procedures. These environments are designed to show how rules shape incentives and how decentralized decisions can generate aggregate outcomes such as prices, allocations, and efficiency measures.

Market and auction experiments are especially valuable because they connect individual decisions to system-level outcomes. Students can observe price formation, gains from trade, allocative efficiency, and the distributional consequences of different mechanisms. They can also compare the performance of institutions under alternative assumptions about information, competition, and valuation.

Implementation and Technical Infrastructure

The technical infrastructure is designed to make the experiments usable in ordinary teaching settings. Students participate through a browser from a mobile phone, laptop, or other web-enabled device. No local installation is required, which reduces the logistical burden of running experiments in large classes.

At the beginning of the course, each student can be assigned or asked to select a private identification code from a predefined list. The same code can then be used across sessions. This procedure preserves anonymity during classroom discussion while allowing the instructor to

link repeated decisions by the same participant across different experiments. It also supports longitudinal analyses of behavior without publicly associating decisions with student identities.

The server-side environment is typically hosted on a controlled university server. This makes it possible to manage access, store data in a predictable location, and comply with institutional requirements for data security. During each session, oTree handles decision submission, matching, payoff calculation, and data aggregation. Results can then be displayed or exported immediately after the experiment, allowing the instructor to base the discussion on data generated by the class.

A recent development in the toolkit is the addition of AI-mediated feedback to selected applications. In the current implementation, the Beauty Contest and Public Goods Game apps can generate a concise instructor-facing commentary from the aggregate data shown in the oTree admin report. The commentary is optional and depends on the availability of the OpenAI Python package and an API key. When enabled, the app sends a structured summary of the session results to the model and requests a short interpretation suitable for classroom debriefing. When the service is unavailable, the experiment continues normally and the report displays an explanatory fallback message.

Classroom Workflow

A typical classroom session follows four steps.

1. **Introduction**: The instructor introduces the decision environment, clarifies the rules, and asks students to form expectations about the likely outcome.
2. **Participation**: Students participate through the web interface, submitting decisions from their own devices while the server records choices and applies the relevant matching or payoff rules.
3. **Outcomes**: The instructor displays aggregate results immediately after the task, such as distributions of choices, winning guesses, transaction prices, efficiency measures, or AI-mediated debriefing notes where available.
4. **Interpretation**: The class uses these results to interpret the theoretical benchmark, identify deviations from it, and discuss how incentives, beliefs, information, and institutional rules shaped behavior. AI-generated comments, when used, act as prompts for discussion rather than as final explanations of behavior.

This workflow makes the experiment part of the explanation rather than a separate demonstration: students move from choice to data, from data to theory, and from theory back to interpretation.

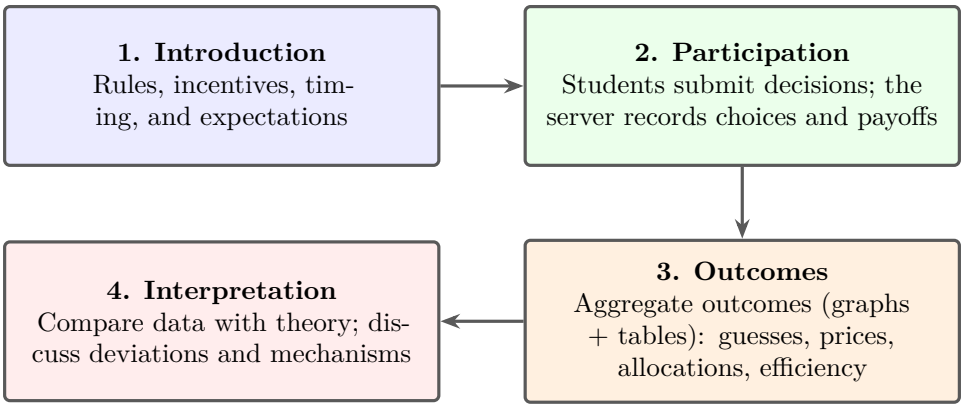


Figure 1: Classroom workflow for an EconLab experiment.

oTree Framework

The applications are implemented using the oTree framework Chen et al. (2016), which is a Python-based platform for building and running online experiments. Full documentation for oTree is available at https://otree.readthedocs.io/en/latest/. oTree is distributed under a MIT-style open-source license, with a mandatory academic citation requirement for published results.

Participants interact with the apps through a web interface, and the server manages the logic of the experiment, including participant matching, decision processing, and data storage. The use of oTree allows for flexibility in designing the experiments, as well as ease of deployment and data management. Each app is structured as a standard oTree application, with defined models, views, and templates. The models define the data structure and logic of the experiment, while the views handle the user interface and interaction. The templates are used to create the HTML pages that participants see when they interact with the app. This modular structure allows for easy customization and adaptation of the apps to different teaching contexts and objectives.

Programming is done in Python, and the user interface is built using HTML, CSS, and JavaScript. Since February 2026, the OTAI platform has made it possible to build oTree apps through an AI-assisted interface, which further reduces the technical barrier for instructors who want to create or modify experiments for their classes.

For classroom use, oTree's main advantage is that the dashboard gives the instructor immediate access to submissions, group outcomes, and exportable data. Its support for multiple languages and customizable templates also makes the same application adaptable to different courses, student populations, and teaching formats.

The AI-feedback layer builds on this dashboard logic. The app first constructs a structured data payload from the choices, payoffs, group statistics, and language setting of the session.

This payload is then used to generate a short HTML fragment for the admin report. The prompts are constrained to describe observed aggregate patterns, to cite numeric values where useful, and to avoid unsupported claims about individual motives. This design keeps the model in a supporting role: it helps the instructor identify patterns and formulate debriefing questions, while the empirical record remains the primary object of discussion.

An example: Beauty Contest Game

The Beauty Contest game illustrates the role of strategic reasoning and beliefs about others' reasoning (Nagel, 1995). Students submit numerical guesses, expressed as integer values in the range 0–100, and the winning guess is determined by a rule based on the average response in the group. Specifically, the winning guess is the number that is closest to a fraction (e.g., 2/3) of the average guess submitted by all participants. The theoretical benchmark for this game is that rational players iteratively eliminate dominated strategies, leading to a unique Nash equilibrium in which all players guess zero. However, in practice, students often submit guesses that are higher than the equilibrium prediction, reflecting different levels of reasoning and beliefs about others' behavior. The activity provides a compact way to introduce iterated reasoning, equilibrium prediction, and deviations from equilibrium play.

Interface and Configuration

The ***init****.py* code in the Appendix illustrates how the Beauty Contest app defines the main parameters of the activity, including the multiplier, the valid range of guesses, the number of rounds, and the language of the interface. The *admin_report* function summarizes individual guesses, group-level outcomes, and the distribution of responses, giving the instructor the material needed for the classroom debriefing. The current version also includes an optional AI-mediated commentary in the admin report. The app packages the average guess, the target value, the best guess, the number of winners, and the distribution of submitted guesses into a structured prompt and asks the model to produce a short interpretation in the language of the session.

The participant interface is deliberately simple: students read the instructions, enter a valid numerical guess, and submit their decision. This design keeps attention on the strategic problem rather than on the mechanics of the web interface.

The following screenshots illustrate the main components of the user interface for the Beauty Contest game. The design emphasizes clarity and ease of use, ensuring that students can focus on the strategic aspects of the game rather than on navigating the interface.

Instructions

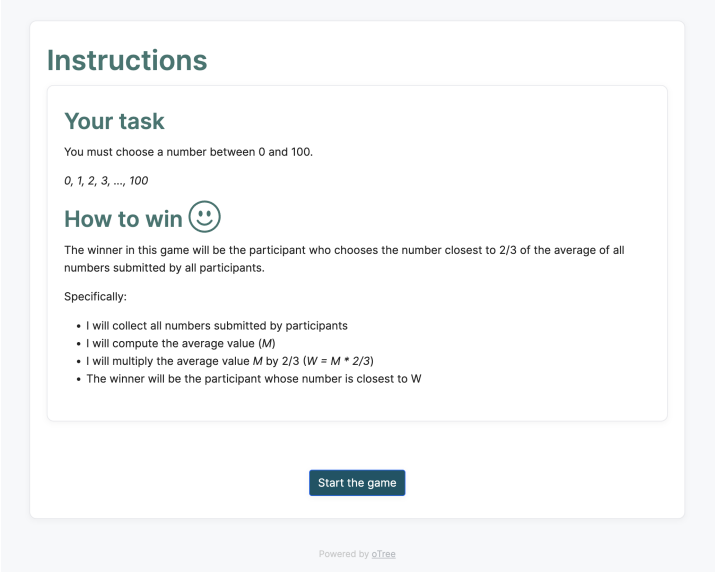


Figure 2: Instructions screen for the Beauty Contest game.

Each participant receives clear instructions about the task, the payoff structure, and the winning condition. The instructions can be customized to include additional information or examples as needed.

Guessing interface

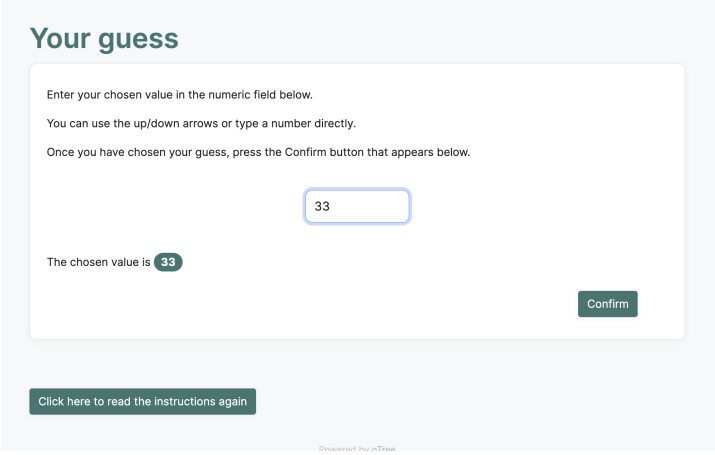


Figure 3: Guessing interface for the Beauty Contest game.

The guessing interface allows participants to enter their guess using a numeric input field. The interface includes validation to ensure that the input is within the allowed range, and it provides immediate feedback on the value entered.

Feedback after submission

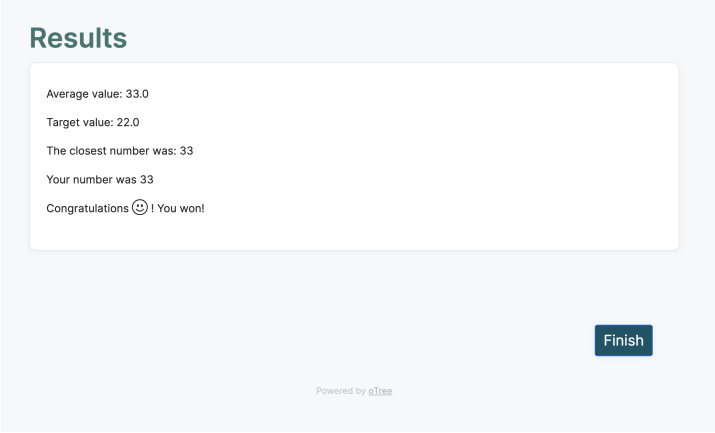


Figure 4: Feedback screen after submitting a guess in the Beauty Contest game.

After all participants have submitted their guesses, the app computes the average guess, applies the winning rule, and displays the results. If any participant fails to submit a guess, the app will automatically exclude the participant from the computation and notify them accordingly. This ensures that the results are based only on valid inputs and maintains the integrity of the experiment.

Results and discussion

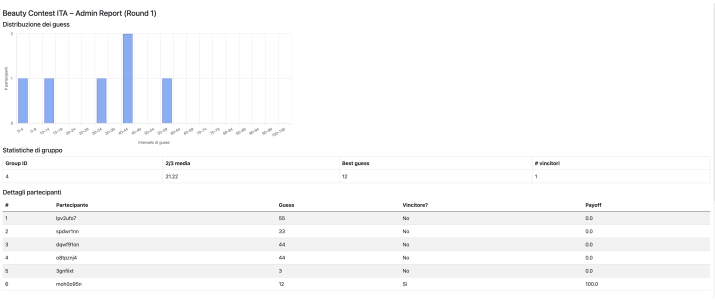


Figure 5: Results screen showing the distribution of guesses in the Beauty Contest game.

The report panel displays the distribution of guesses, the average guess, and the winning guess in real time. This allows the instructor to facilitate a discussion about strategic reasoning, equilibrium predictions, and observed behavior in class.

When AI-mediated feedback is enabled, the same report can also display a brief interpretive comment for the instructor. In the Beauty Contest app, the comment focuses on the distribution of guesses, the relationship between the average guess and the target value, the location of the winning guess, and possible debriefing questions about iterated reasoning. In the Public

Goods Game app, the corresponding feedback summarizes cooperation levels, round-to-round dynamics, differences between groups, and the relationship between contributions and payoffs.

Results and Data Analysis

Figure 6 shows the distribution of guesses across a series of classroom sessions using the Beauty Contest game. The graph illustrates how students' responses cluster around certain values and how they deviate from the equilibrium prediction.

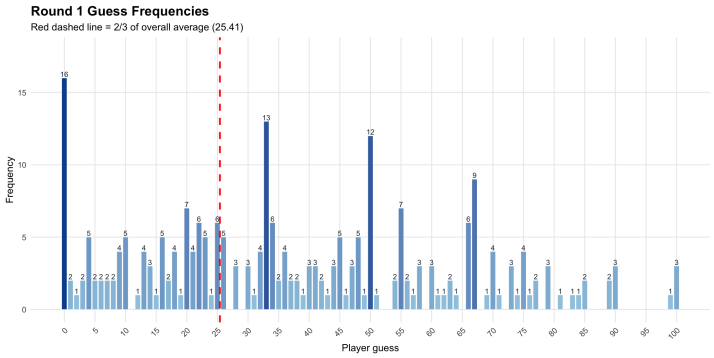


Figure 6: Distribution of guesses in the Beauty Contest game.

Interestingly, the data show spikes at the values corresponding to k-levels 0 (=50) and 1 (=33), as well as at the Nash equilibrium prediction (=0). When data are presented to participants, these patterns can be used to discuss the concept of iterated reasoning and how it shapes strategic behavior. The presence of spikes at certain values suggests that many participants are using similar levels of reasoning, which can be a starting point for discussing how beliefs about others' behavior influence one's own decisions. Moreover, it is possible to link the observed distribution to the "paradox of the beauty contest": the equilibrium prediction is counterintuitive, is not the most common response in practice, and is generally not the guess closest to the winning guess. This can be used to motivate a discussion about the limitations of equilibrium analysis and the importance of behavioral factors in strategic settings.

Repository and License

The applications are distributed through a public GitHub repository so that they can be inspected, downloaded, adapted, and deployed using standard oTree procedures. The project is released under the Creative Commons Attribution 4.0 International license, which permits reuse, modification, and redistribution provided that appropriate attribution is given.

The repository is intended to support transparency, replicability, and collaborative development of teaching-oriented experimental tools. Each app is accompanied by a `README.txt` file

describing the economic environment implemented, the theoretical background, the timing and structure of the task, the main parameters, configuration options, and suggested classroom uses. This documentation is meant to make the apps easier to adopt and adapt across different courses and institutional contexts.

Conclusion

EconLab Classroom shows how browser-based experiments can be used to make economic theory observable in ordinary teaching settings. Across modules on preferences, strategic interaction, and market institutions, the common contribution is not only the availability of individual apps but the workflow they support: students make decisions, the class observes the resulting data, and the instructor uses those outcomes to revisit the theoretical benchmark.

The Beauty Contest example illustrates this logic in a compact setting. It links a clear equilibrium prediction to a distribution of classroom responses, making deviations from the benchmark visible and available for discussion. The recent addition of AI-mediated feedback extends this logic by helping the instructor convert session-level data into concise discussion prompts, while preserving the central role of the observed classroom data. More generally, the project provides instructors with adaptable tools for turning classroom sessions into empirical exercises in which models, behavior, and institutions can be examined together.

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Appendix

Code for the Beauty Contest Game

init.py